

Define Problem Statements

1. Problem Statement Definition

Localized areas face sudden flood occurrences during severe monsoons. Traditional forecasting tools are slow, require massive computational power, and are not easily accessible by ground-level relief coordinators.

- The Goal: Provide a lightweight, zero-latency, high-accuracy classification system that runs on any device (including mobile) to evaluate flood susceptibility instantaneously.

2. Hydrological Danger Baselines

- Safe Threshold: Combined seasonal rainfall under historical average baselines.
- Danger Threshold: June-September rainfall exceeding 2425.64 mm represents anomalous hydrological load, triggering flood alarms.